



I BUILT A RAG THAT CAN ACTUALLY SEE PDFs

Most RAG systems only read text.

- ✗ Misses tables buried in scans
- ✗ Ignores charts and graphs entirely
- ✗ Loses the document's visual layout

Vision RAG understands PDFs visually — like a human reader.

SWIPE



Traditional RAG isn't reading your PDF. It's reading extracted text.

When a PDF contains:



Charts



Tables



Graphs



Complex layouts

*Important information gets lost
before retrieval even begins.*

// WHY THIS MATTERS

Imagine asking:

“Which quarter had the highest revenue?”

The answer might exist inside a chart.



HUMANS

Instantly understand visual information.



MOST RAG

Systems simply don't.

I built a Vision RAG.

Instead of treating PDFs as text, it treats every page as a visual document.

THE MODEL UNDERSTANDS:

✓ Text

✓ Tables

✓ Charts

✓ Layout

✓ Visual relationships

Vision RAG sees the page the way a person would — layout, charts, and text together.

From raw PDF to a grounded answer.



PDF

// Raw document, untouched



Convert pages into images

// Every page becomes a visual



ColQwen2 generates visual embeddings

// Layout, charts, and text encoded together



Store embeddings in Qdrant

// Fast, scalable vector search



Retrieve relevant pages

// Find the pages that matter



LLM generates answers

// Grounded in what was actually seen

Five tools. One seamless pipeline.



ColQwen2

// Visual embedding model



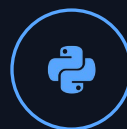
Qdrant

// Vector database



OpenAI

// Answer generation



Python

// Core pipeline



Streamlit

// Interactive UI

Built end-to-end for visual document understanding.

TRADITIONAL RAG ASKS

“What words are present?”

VS

VISION RAG ASKS

“What information is visible?”

That's the difference between reading text and understanding documents.



// WHAT IT CAN UNDERSTAND



Business Reports



Financial Statements



Performance Dashboards



Research Papers



Multi-column Docs



Technical PDFs



College Textbooks



Lecture Notes



Study Materials

// WHO CAN BENEFIT?



Students



Researchers



Business Analysts



Educators



Data Teams



Anyone with large PDF collections

Find answers in seconds — not hundreds of pages.



Thanks for reading.

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